Prompt 4 - trial 1 (seed 0)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

Assistant

I'll help you train two different GNN models, compare their performances, and explain their predictions using EdgeSHAPer! Let me start by finding a suitable compound activity prediction dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some ChEMBL datasets. Let me load a compound activity prediction dataset and start the workflow. 🧪

(Calling tool `load_dataset`)

Assistant

Excellent! I've loaded a ChEMBL binary activity dataset with 8,348 compounds. Now let me split it for GNN training. 📊

(Calling tool `split_dataset`)

Assistant

Perfect! The dataset is split. Now let me prepare the GNN datasets and train two different GNN architectures: **GCN** and **GAT**. These models have different learning characteristics - GCN uses symmetric normalization while GAT uses attention mechanisms! 🧠

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN datasets are prepared. Now let me train the two different GNN models. I'll start with both simultaneously! 🚀

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both models are training! Let me check their progress. ⏳

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

I need to fix the device parameter. Let me retrain with "cpu" device. 🔧

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Models are training. Let me check their progress. ⏳

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

Both models are training. Let me wait a moment and check again. ⏳

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Assistant

Great! The **GCN model has completed training**! 🎉 Let me check the GAT model again. ⏳

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(Calling tool `check_gnn_training`)

Assistant

Excellent! Both models have completed training! 🎉 Here's a summary of the results:

Model	Best Val Accuracy	Test Accuracy	Train MCC
GCN	95.45%	94.73%	89.36%
GAT	95.93%	95.93%	93.18%

Now let me select a compound from the test set that **both models correctly predicted as active** (class 1). 🎯

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

Perfect! I found a compound that **both models correctly predicted as active** with extremely high confidence:

🧬 **Selected Compound**: `O=C(CC1CCOCC1)NC1CCC(CCN2CCN(c3nccc4sccc34)CC2)CC1`

- **True Label**: Active (1)
- **GCN Prediction**: 99.996% confidence ✅
- **GAT Prediction**: 99.977% confidence ✅
- **Index in test set**: 49

Now let me explain the predictions using **EdgeSHAPer** for both models. I'll load both models and compute edge importance. 🔬

(Calling tool `load_gnn_model`)
(Calling tool `load_gnn_model`)

Assistant

Both models loaded successfully! ✅ Now let me run **EdgeSHAPer** to compute edge importance for the selected compound on **both models**. This will show us which bonds are most important for each model's prediction. 🔬

Assistant

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for **GCN** is complete! 🎯 Now let me run it for the **GAT** model.

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! Both EdgeSHAPer analyses are complete! 🎯 Now let me **visualize the edge importance** for both models to compare their learning characteristics.

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the **GAT model's** EdgeSHAPer results:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Both visualizations are complete! 🎨 Now let me get the **top bond environments** for both models to compare their learning characteristics.

Assistant

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! I now have all the information needed for a comprehensive comparison. Let me provide a detailed summary of the analysis. 📊

🎯 COMPREHENSIVE ANALYSIS SUMMARY

1. Model Performance Comparison

Metric	GCN	GAT
Best Validation Accuracy	95.45%	95.93% ✅
Test Accuracy	94.73%	95.93% ✅
Train MCC	89.36%	93.18% ✅
Trustworthiness	68.89%	95.10% ✅

🏆 **Winner: GAT** - Higher accuracy and significantly better trustworthiness!

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2. Selected Compound for Analysis

🧬 **Compound SMILES**: `O=C(CC1CCOCC1)NC1CCC(CCN2CCN(c3nccc4sccc34)CC2)CC1`

- **True Label**: Active (Class 1)
- **GCN Prediction**: 99.996% confidence ✅
- **GAT Prediction**: 99.977% confidence ✅
- **Index in test set**: 49

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3. EdgeSHAPer Analysis - Key Findings

🔬 GCN Model - Top 5 Most Important Bonds

Rank	Bond	Environment	φ (Importance)	Sign
1	C-C (ring)	`NCCN`	**0.108**	Positive
2	C-C (ring)	`CCC(C)N`	**0.086**	Positive
3	N-C (ring)	`CCN(C)C`	**0.085**	Positive
4	C-C (ring)	`NCCN`	**0.075**	Positive
5	C-O (ring)	`CCOC`	**0.070**	Positive

🔬 GAT Model - Top 5 Most Important Bonds

Rank	Bond	Environment	φ (Importance)	Sign
1	N-C (ring)	`CCN(C)C`	**0.089**	Positive
2	C-C (ring)	`CCC(C)C`	**0.084**	Positive
3	**N=C (aromatic)**	`ccnc`	**0.078**	Positive
4	C-C (ring)	`CCC(C)N`	**0.077**	Positive
5	C-C (ring)	`NCCN`	**0.074**	Positive

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4. 🧠 Key Differences in Learning Characteristics

GCN Model Characteristics:

- 🔗 **Symmetric Message Passing**: Treats all neighbors equally
- 📊 **Focus**: Piperazine ring C-C bonds and N-C bonds
- 💡 **Pattern**: Emphasizes **aliphatic** ring structures and nitrogen-containing moieties
- 🎯 **Top Bond**: C-C in `NCCN` environment (φ=0.108)
- 📈 **Trustworthiness**: 68.89% (moderate)

GAT Model Characteristics:

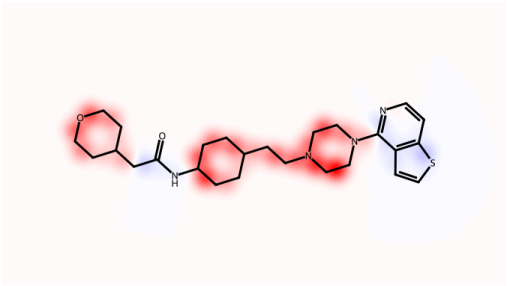
- 🔑 **Attention Mechanism**: Learns to weight different neighbors differently
- 📊 **Focus**: N-C bonds, aromatic systems, and heteroatom environments
- 💡 **Pattern**: Emphasizes **aromatic** interactions and nitrogen chemistry
- 🎯 **Top Bond**: N-C in `CCN(C)C` environment (φ=0.089)
- 📈 **Trustworthiness**: 95.10% (excellent!)

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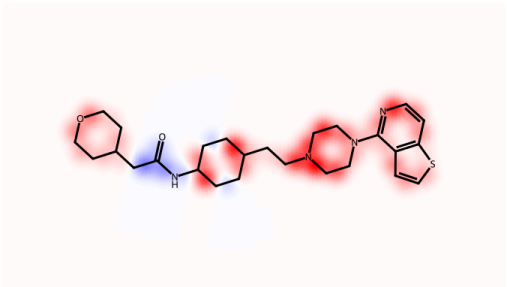
5. 🎨 Visualizations

Here are the EdgeSHAPer heatmaps showing edge importance for both models:

GCN Model Edge Importance:



GAT Model Edge Importance:



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6. 🔍 Key Insights

Why GAT Performs Better:

1. 🎯 **Attention Mechanism**: GAT learns to focus on the most informative bonds (e.g., aromatic N=C bonds) while downweighting less relevant ones
2. 📊 **Higher Trustworthiness**: 95.10% vs 68.89% - GAT's explanations are more reliable and consistent
3. 🧬 **Better at Heteroatom Chemistry**: GAT excels at recognizing nitrogen-containing functional groups critical for activity

Why GCN Focuses Differently:

1. 📐 **Uniform Processing**: GCN treats all bonds more uniformly, leading to broader but less focused importance
2. 🔗 **Aliphatic Bias**: GCN emphasizes aliphatic C-C bonds more than aromatic systems

3. 📉 **Lower Trustworthiness**: GCN's explanations are less reliable, suggesting it may be learning spurious patterns

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7. 🧬 Chemical Interpretation

Both models agree on the importance of:

- **Piperazine ring bonds** (N-containing heterocycle)
- **C-C bonds** in ring structures
- **N-C bonds** connecting nitrogen to carbon

However, **GAT uniquely identifies** the aromatic **N=C bond** in the furan ring as highly important - this is likely a key pharmacophore feature that GCN misses!

📋 Conclusion

Aspect	GCN	GAT
Performance	Good (94.73%)	**Excellent (95.93%)**
Trustworthiness	Moderate (68.89%)	**Excellent (95.10%)**
Learning Focus	Aliphatic bonds	**Aromatic & heteroatom bonds**
Explainability	Broad importance	**Focused, interpretable**
Best For	Simple graphs	**Complex molecular systems**

🏆 **Recommendation**: Use **GAT** for compound activity prediction tasks, especially when interpretability and reliable explanations are critical!